Mitsubishi MITAJE71 protocol driver

Overview

This driver supports the A Series of PLC's. Revision 103 onwards also supports the the FX3U PLC.

NOTE :

Support for QnA and Q series controllers has moved to the Mitsubishi Q driver, revision 101 no longer supports QnA or Q processors.

The SNMP and FTP services are not required for *Adroit* to Mitsubishi communication. For improved network performance and reliability, it is recommended that you install 2 network adapters into your PC. Use the first adapter for *Adroit* to *Adroit* and all other NT connectivity. The first adapter may have all the default installation bindings. Use the second adapter for *Adroit* to PLC (AJ71E71 communication processor) connectivity and bind only to the TCP/IP protocol.

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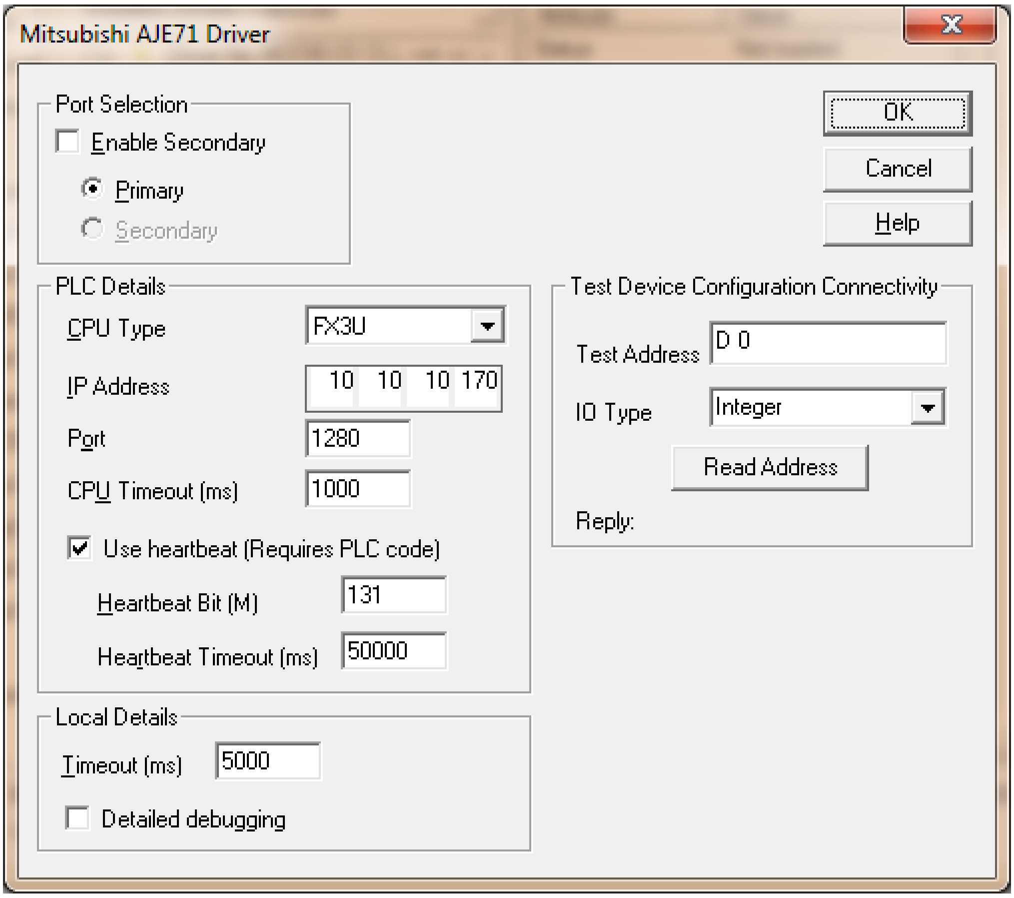
Device Configuration

One device agent must be configured for each CPU to be communicated to.



PLC Details

CPU Type - The following CPU types are supported; A1, A1N, A2, A2N, A2C, AOJ2H, A2-S1, A2N-S1, A3N, A3M, A2A, A2A-S1, A3, Q-Series,Fx3U.

IP Address - Identifies the Mitsubishi PLC's IP addresses. This address must be unique. It defaults to your local IP (40.0.0.1).

Port - TCP/IP port number. The PLC software shipped with the driver is programmed for port 1280 - 1287.

Heartbeat Bit (M) - In order to guarantee the re-establishment (under all circumstances) of a connection between *Adroit* and the Mitsubishi PLC, the heartbeat bit is used. *Adroit* will set this bit every Heartbeat time-out period. The PLC program will reset this bit immediately after *Adroit* has set it. The PLC program will also monitor this bit and reset the connection if the 120 seconds (PLC program default) has elapsed without being set by *Adroit*.

Heartbeat Time-out - The heartbeat pulse generated by *Adroit* to keep the communication link alive. *Adroit* will set the Heartbeat Bit every Heartbeat time period.

CPU Time-out - This is the time-out period for transactions between the AJ71E71 card and the PLC CPU.

Local Details

Time-out - The maximum time allowed for *Adroit* to PLC communication messages before a time-out error will be registered.

Detailed debugging - Currently detailed debugging does the following:

1. Shows transaction start and end times
2. Describes each read opperation before executing it

The Agent server must as always be re-started for the options changed in this dialog to take effect.

Test Device Config for Connectivity

This section is available to help users determine whether or not the settings they have entered are valid and Adroit can communicate with your device.

Test Address – An address in your device which will be polled when the “Read Address” button is clicked. The address entered must be valid.

IO Type – The address type being read. ie Boolean, integer, real or string.

Read Address – Click this button to poll your device with the current settings in your dialog.

Reply – When “Read Address” is clicked, Adroit will poll your device. The value returned from your device will be displayed here.

Wiring

As per Ethernet standards.

AJ71E71 module

Set the Mode switch to 0.

Set SW-7 to ON to allow *Adroit* to write to PLC memory.

All other switches must be off.

Ref: Mitsubishi Electric Corporation AJ71E71 User Manual, Document Number IB (NA) 66310-A (9105) MEE, page 4-6, May 1991

AJ71E71N-B5T

Note: The dip-switch SW-7 is on the mini-dip inside the module.

PLC software

A Series PLC setup

PLC software is required to allow communication with the AJ71E71 card. Copies of this software for the Mitsubishi AJ71E71 card is available as part of the *Adroit* 3.51 distribution in the Samples/MitPLC sub-directory. A GX-Developer 7 project is also available.

All PLC items used by this software has to be checked for possible conflict with your existing PLC program.

Fx3U Series PLC setup

FX Configurator-EN software is required to allow communication with the Fx3U Ethernet card. This software is available from CBI Automation.

Appendix B has step by step instructions on how to setup the Ethernet card that Adroit can talk to it using the FX Configurator-EN software.

Software Note - Copies of the Mitsubishi PLC software for the AJ71E71 adapter is available on a *No Guarantee* basis and the standard terms of *Adroit* software distribution apply. Adroit Technologies will therefore not accept any responsibility for any direct, indirect or consequential damages of any nature that may occur as a result of the use of such software.

A listing of the program follows appears in appendix A.

Troubleshooting

It is possible that by modifying the default settings, that the **watchdog** sent by the driver may not arrive in the PLC, thus causing the PLC to “intermittently” reset our connection to the PLC. Ensure that the timer in the PLC (which runs for 120 seconds by default) is at least double the setting used by the driver. If the driver channel is backlogged (ie the driver is too busy with polling tasks), the watchdog may also be delayed,as it is not a foreground process in the driver. Driver backlogging or queing is easily detectable by setting a value in register in the PLC, and then timing how long it takes to receive a feedback via another tag which was scanned to the same register.

Supported addresses

Mitsubishi A series PLC data (device) types supported in the *Adroit* Protocol Driver Product Version 3.51 Build 151 (Pvt. Build 88)

Please see Table 8.2 on page 8-15 of the Ethernet Interface Module Manual : IB (NA) 66310-A (9105) MEE

Device		Min Range	Max Range*	PLC Data Type	BOOLEAN	INTEGER	REAL	STRING
Data Reg	D	0	6143	Int 16	D 1.2	D 0	D 0	D 0 Sn
			6143	BCD 16		D 0 B	D 0 B	
			6142	BCD 32		D 0 C	D 0 C	
			6142	Long		D 0 L	D 0 L	
			6142	Real 32			D 0 R	
Data Reg	D	9000	9255	Int 16	D 1.2	D 0	D 0	D 0 Sn
			9255	BCD 16		D 0 B	D 0 B	
			9254	BCD 32		D 0 C	D 0 C	
			9254	Long		D 0 L	D 0 L	
			9254	Real 32			D 0 R	
Link Reg	W	0	0xFFF	Int 16	W1.2	W 0	W 0	W 0 Sn
File Reg	R	0	8191	Int 16	R 0.0	R 0	R 0	R 0 Sn
			8191	BCD 16		R 0 B	R 0 B	
			8190	BCD 32		R 0 C	R 0 C	
			8190	Long		R 0 L	R 0 L	
			8190	Real 32			R 0 R	
Ext File Reg	E	0:0	28:8191	Int 16	E 1:0.0	E 1:0	E 1:0	E 1:0 Sn
			28:8191	BCD 16		E 1:0 B	E 1:0 B	
			28:8190	BCD 32		E 1:0 C	E 1:0 C	
			28:8190	Long		E 1:0 L	E 1:0 L	
			28:8190	Real 32			E 1:0 R	
Timer	T	0	2047	Int 16		T 0	T 0	
Counter	C	0	1023	Int 16		C 0	C 0	
Inputs	X	0	0x7FF	Bool	X 0	X 0 (16)	X 0 (16)	
			0x7F0	BCD 16		X 0 B	X 0 B	
Outputs	Y	0	0x7FF	Bool	Y 0			
Internal Relay	M/L/S	0	8191	Bool	M 0			
		9000	9255	Bool	L 0			
Link Relay	B	0	0xFFF	Bool	B 0			
Annunciator	F	0	0x7FF	Bool	F 0			

Address register numbers are in decimal or HEX notation depending on:

Device	Prefix	Notation
Data Reg	D	DECIMAL
Link Reg	W	HEX
File Reg	R	DECIMAL
Ext File Reg	E	DECIMAL
Timer	T	DECIMAL
Counter	C	DECIMAL
Inputs	X	HEX
Outputs	Y	HEX
Internal Relay	M,L,S	DECIMAL
Link Relay	B	HEX
Annunciator	F	DECIMAL

Mitsubishi Fx3U series PLC data (device) types supported in the *Adroit* Protocol Driver Product Version 3.51 Build 151 (Pvt. Build 88)

The FX3U PLC is supported from revision 103 of the MITAJE71 driver.

Please note:

1. Only CN and TN registers are supported by using the C and T address type codes respectively. CS and TS registers are not supported at this time
2. There is not support for the M8000 to M8511 range of registers at this time

Please see Table on page 9-17 of the Ethernet Interface Module Manual

Device		Min Range	Max Range*	PLC Data Type	BOOLEAN	INTEGER	REAL	STRING
Data Reg	D	0	7999	Int 16	D 1.2	D 0	D 0	D 0 Sn
			7999	BCD 16		D 0 B	D 0 B	
			7998	BCD 32		D 0 C	D 0 C	
			7998	Long		D 0 L	D 0 L	
			7998	Real 32			D 0 R	
Data Reg	D	8000	8511	Int 16	D 1.2	D 0	D 0	D 0 Sn
			8511	BCD 16		D 0 B	D 0 B	
			8510	BCD 32		D 0 C	D 0 C	
			8510	Long		D 0 L	D 0 L	
			8510	Real 32			D 0 R	
File Reg	R	0	32767	Int 16	R 0.0	R 0	R 0	R 0 Sn
			32767	BCD 16		R 0 B	R 0 B	
			32766	BCD 32		R 0 C	R 0 C	
			32766	Long		R 0 L	R 0 L	
			32766	Real 32			R 0 R	
Timer (TN only)	T	0	511	Int 16		T 0	T 0	
Counter (CN only)	C	0	255	Int 16		C 0	C 0	
Inputs	X	0	377	Bool	X 0	X 0 (16)	X 0 (16)	
			377	BCD 16		X 0 B	X 0 B	
Outputs	Y	0	377	Bool	Y 0			
State	S	0	4095	Bool	S 0			
Internal Relay	M	0	7679	Bool	M 0			
Not supported	M	8000	8511	Bool	M 0			

Address register numbers are in decimal or HEX notation depending on:

Device	Prefix	Notation
Data Reg	D	DECIMAL
Link Reg	W	HEX
File Reg	R	DECIMAL
Ext File Reg	E	DECIMAL
Timer	T	DECIMAL
Counter	C	DECIMAL
Inputs	X	HEX
Outputs	Y	HEX
Internal Relay	M,L,S	DECIMAL
Link Relay	B	HEX
Annunciator	F	DECIMAL

Message protocol

Data Registers

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 44	00	00

Read response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	ff	0a 00	00 00 00 00	20 44	00	00

First word	Last Word
00 00	00 00

Write response:

Subheader	Completion Code
83	00

Link Registers

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 57	00	00

Read response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	Ff	0a 00	00 00 00 00	20 57	00	00

First word	Last Word
00 00	00 00

Write response:

Subheader	Completion Code
83	00

File Registers

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 52	00	00

Read response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	ff	0a 00	00 00 00 00	20 52	00	00

Write response:

Subheader	Completion Code
83	00

Extended File Registers

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Block	Length	End
17	ff	0a 00	00 00 00 00	20 52	01 00	00	00

Read response:

Subheader	Completion Code	First data word	Last data word
97	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Block	Length	End	First word	Last Word
18	ff	0a 00	00 00 00 00	20 52	01 00	00	00	00 00	00 00

Write response:

Subheader	Completion Code
98	00

Timers

Read request (Present Value):

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	4e 54	00	00

Response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End	First word	Last Word
03	ff	0a 00	00 00 00 00	4e 54	00	00	00 00	00 00

Write response:

Subheader	Completion Code
83	00

Counters

Read request (Present Value):

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	4e 43	00	00

Response:

Subheader	Completion Code	First data word	Last data word
81	00	00 00	00 00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
03	ff	0a 00	00 00 00 00	4e 43	00	00

First word	Last Word
00 00	00 00

Write response:

Subheader	Completion Code
83	00

Inputs

Read request:

Read bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
00	ff	0a 00	00 00 00 00	20 58	00	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
80	00	00	00

Write request:

Write word	PC CPU	2500 ms t/o	Offset	Type	Length	End
02	ff	0a 00	00 00 00 00	20 58	00	00

First byte	Last byte
00	00

Write response:

Subheader	Completion Code
82	00

Outputs

Read request:

Read bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
00	ff	0a 00	00 00 00 00	20 59	00	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
80	00	00	00

Write request:

Write bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
02	ff	0a 00	00 00 00 00	20 59	00	00

First byte	Last byte
00	00

Write response:

Subheader	Completion Code
82	00

Internal Relay

Read request:

Read bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
00	ff	0a 00	00 00 00 00	20 4d	00	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
80	00	00	00

Write request:

Write bit	PC CPU	2500 ms t/o	Offset	Type	Length	End
02	ff	0a 00	00 00 00 00	20 4d	00	00

First byte	Last byte
00	00

Write response:

Subheader	Completion Code
82	00

Link Relay

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 42	01	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
81	00	00	00

Write request:

Write bit	PC CPU	2500 ms t/o	Offset	Type	Length
02	ff	0a 00	00 00 00 00	20 42	01

First byte	Last byte	End
00	00	00

Write response:

Subheader	Completion Code
82	00

Annunciators

Read request:

Read word	PC CPU	2500 ms t/o	Offset	Type	Length	End
01	ff	0a 00	00 00 00 00	20 46	01	00

Read response:

Subheader	Completion Code	First data byte	Last data byte
81	00	00	00

Write request:

Write bit	PC CPU	2500 ms t/o	Offset	Type	Length
02	ff	0a 00	00 00 00 00	20 46	01

First byte	Last byte	End
00	00	00

Write response:

Subheader	Completion Code
82	00

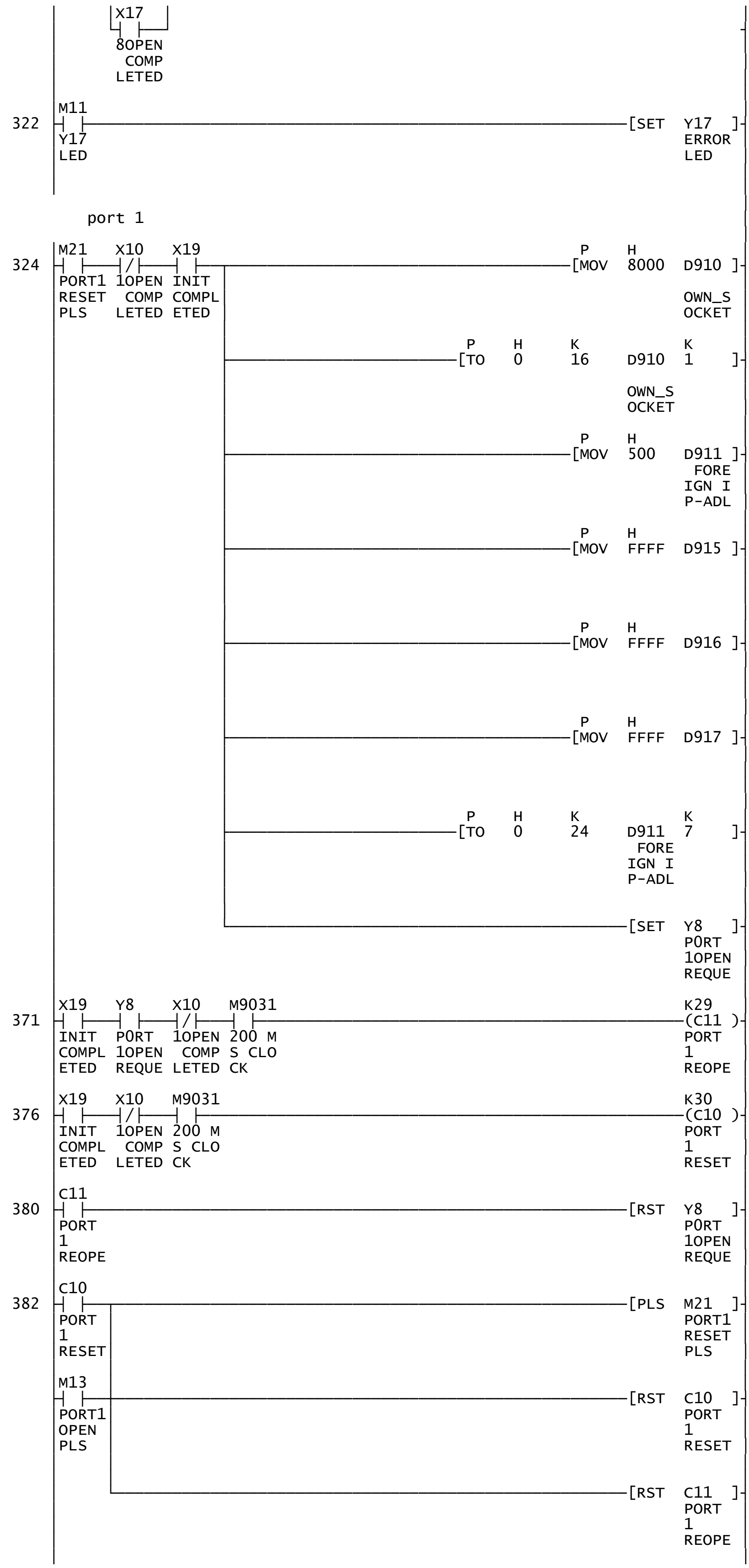
Appendix A

```
****************************************************
*                   COMS START                     *
****************************************************

IP Address of 40.0.0.5  and Port of 1280

    M9038                                                H
 0  ┤ ├──┬──────────────────────────────────────[DMOV   28000005    D900 ]
    ONLY │                                                          LOW
    1 SCA│                                                          IP-AD
    N ON │                                                          DRESS
         │                                       H      K           K
         ├────────────────────────────────[TO    0      0    D900   2     ]
         │                                                   LOW
         │                                                   IP-AD
         │                                                   DRESS
         │                                                   K
         ├──────────────────────────────────────────[MOV     2      D902 ]
         │                                                          TCP T
         │                                                          IMEOU
         │                                                          T
         │                                                   K
         ├──────────────────────────────────────────[MOV     1      D903 ]
         │                                                          TCP Z
         │                                                          ERO W
         │                                                          INDOW
         │                                                   K
         ├──────────────────────────────────────────[MOV     1      D904 ]
         │                                                          TCP R
         │                                                          ETRAN
         │                                                          S.TIM
         │                                                   K
         ├──────────────────────────────────────────[MOV     1      D905 ]
         │                                                          TCP C
         │                                                          OMPLE
         │                                                          T.TIM
         │                                                   K
         ├──────────────────────────────────────────[MOV     2      D906 ]
         │                                                          IP AS
         │                                                          SEMBL
         │                                                          Y TIM
         │                                                   K
         ├──────────────────────────────────────────[MOV     2      D907 ]
         │                                                          RESP.
         │                                                           WATC
         │                                                          HDOG
         │                                       H      K           K
         ├────────────────────────────────[TO    0      10   D902   6     ]
         │                                                   TCP T
         │                                                   IMEOU
         │                                                   T
         │
         └──────────────────────────────────────────────────[SET    Y19  ]
                                                                    MODUL
                                                                    EINIT
                                                                    REQUE

       Module ini error

    X1A                                          H      K           K
57  ┤ ├──┬────────────────────────────────[FROM  0      80   D990   1     ]
    MODUL│
    INIT │
    ERROR│
         ├──────────────────────────────────────────────────[RST    Y19  ]
         │                                                          MODUL
         │                                                          EINIT
         │                                                          REQUE
         │
         └──────────────────────────────────────────────────[PLF    M10  ]
                                                                    INI
                                                                    ERROR
                                                                    PLF
    M10
71  ┤ ├─────────────────────────────────────────────────────[SET    Y17  ]
    INI                                                             ERROR
    ERROR                                                           LED
    PLF
```

```
                                                     H      K      K      K
       ├─────────────────────────────────────────[TO    0      80     0      1      ]-
       │
       │
       │
       └────────────────────────────────────────────────────────────[RST   Y17   ]-
                                                                          ERROR
                                                                          LED

Open connection error
     X18
 83 ─┤ ├──┬──────────────────────────────────────[FROM  H      K             K
                                                        0      93     D991   1      ]-
     OPEN │                                                           PORT1
     ERROR│                                                           ERROR
          │                                                           CODE
          │                                             H      K             K
          ├──────────────────────────────────────[FROM  0      103    D992   1      ]-
          │                                                           PORT3
          │                                                           ERROR
          │                                                           CODE
          │                                             H      K             K
          ├──────────────────────────────────────[FROM  0      113    D993   1      ]-
          │
          │                                             H      K             K
          ├──────────────────────────────────────[FROM  0      123    D994   1      ]-
          │                                                           PORT4
          │                                                           ERROR
          │                                                           CODE
          │                                             H      K             K
          ├──────────────────────────────────────[FROM  0      133    D995   1      ]-
          │                                                           PORT5
          │                                                           ERROR
          │                                                           CODE
          │                                             H      K             K
          ├──────────────────────────────────────[FROM  0      143    D996   1      ]-
          │                                                           PORT6
          │                                                           ERROR
          │                                                           CODE
          │                                             H      K             K
          ├──────────────────────────────────────[FROM  0      153    D997   1      ]-
          │                                                           PORT7
          │                                                           ERROR
          │                                                           CODE
          │                                             H      K             K
          └──────────────────────────────────────[FROM  0      163    D998   1      ]-
                                                                      PORT8
                                                                      ERROR
                                                                      CODE
     X18    X10                                                K
156 ─┤/├────┤ ├──────────────────────────────────────────[MOV   0      D991 ]-
     OPEN   1OPEN                                                     PORT1
     ERROR   COMP                                                     ERROR
            LETED                                                     CODE
     X18    X11                                                K
163 ─┤/├────┤ ├──────────────────────────────────────────[MOV   0      D992 ]-
     OPEN   2OPEN                                                     PORT3
     ERROR   COMP                                                     ERROR
            LETED                                                     CODE
     X18    X12                                                K
170 ─┤/├────┤ ├──────────────────────────────────────────[MOV   0      D993 ]-
     OPEN   3OPEN
     ERROR   COMP
            LETED
     X18    X13                                                K
177 ─┤/├────┤ ├──────────────────────────────────────────[MOV   0      D994 ]-
     OPEN   4OPEN                                                     PORT4
     ERROR   COMP                                                     ERROR
            LETED                                                     CODE
     X18    X14                                                K
184 ─┤/├────┤ ├──────────────────────────────────────────[MOV   0      D995 ]-
     OPEN   5OPEN                                                     PORT5
     ERROR   COMP                                                     ERROR
            LETED                                                     CODE
     X18    X15                                                K
191 ─┤/├────┤ ├──────────────────────────────────────────[MOV   0      D996 ]-
     OPEN   6OPEN                                                     PORT6
     ERROR   COMP                                                     ERROR
            LETED                                                     CODE
```

```
     X18    X16                                                   K
198 -|/|----| |----------------------------------------------[MOV 0     D997 ]-
     OPEN   7OPEN                                                       PORT7
     ERROR   COMP                                                       ERROR
            LETED                                                       CODE

     X18    X17                                                   K
205 -|/|----| |----------------------------------------------[MOV 0     D998 ]-
     OPEN   8OPEN                                                       PORT8
     ERROR   COMP                                                       ERROR
            LETED                                                       CODE

     X10
212 -|/|------------------------------------------------------------[PLS  M13  ]-
     1OPEN                                                               PORT1
      COMP                                                               OPEN
     LETED                                                               PLS

     X11
216 -|/|------------------------------------------------------------[PLS  M14  ]-
     2OPEN                                                               PORT2
      COMP                                                               OPEN
     LETED                                                               PLS

     X12
220 -|/|------------------------------------------------------------[PLS  M15  ]-
     3OPEN                                                               PORT3
      COMP                                                               OPEN
     LETED                                                               PLS

     X13
224 -|/|------------------------------------------------------------[PLS  M16  ]-
     4OPEN                                                               PORT4
      COMP                                                               OPEN
     LETED                                                               PLS

     X14
228 -|/|------------------------------------------------------------[PLS  M17  ]-
     5OPEN                                                               PORT5
      COMP                                                               OPEN
     LETED                                                               PLS

     X15
232 -|/|------------------------------------------------------------[PLS  M18  ]-
     6OPEN                                                               PORT6
      COMP                                                               OPEN
     LETED                                                               PLS

     X16
236 -|/|------------------------------------------------------------[PLS  M19  ]-
     7OPEN                                                               PORT7
      COMP                                                               OPEN
     LETED                                                               PLS

     X17
240 -|/|------------------------------------------------------------[PLS  M20  ]-
     8OPEN                                                               PORT8
      COMP                                                               OPEN
     LETED                                                               PLS

          K             X18
244 -[<>  0     D991 ]-| |----+-------------------------------------[RST  Y8   ]-
                 PORT1 OPEN   |                                          PORT
                 ERROR ERROR  |                                          1OPEN
                 CODE         |                                          REQUE
                              |
     M13                      |
    -| |----------------------+
     PORT1
     OPEN
     PLS

          K             X18
252 -[<>  0     D992 ]-| |----+-------------------------------------[RST  Y9   ]-
                 PORT3 OPEN   |                                          PORT
                 ERROR ERROR  |                                          2 OPN
                 CODE         |                                          REQUE
                              |
     M14                      |
    -| |----------------------+
     PORT2
     OPEN
     PLS

          K             X18
260 -[<>  0     D993 ]-| |----+-------------------------------------[RST  YA   ]-
                       OPEN   |                                          PORT
                       ERROR  |                                          3 OPN
                              |                                          REQUE
                              |
     M15                      |
    -| |----------------------+
     PORT3
     OPEN
     PLS

          K             X18
268 -[<>  0     D994 ]-| |----+-------------------------------------[RST  YB   ]-
                 PORT4 OPEN   |                                          PORT
```

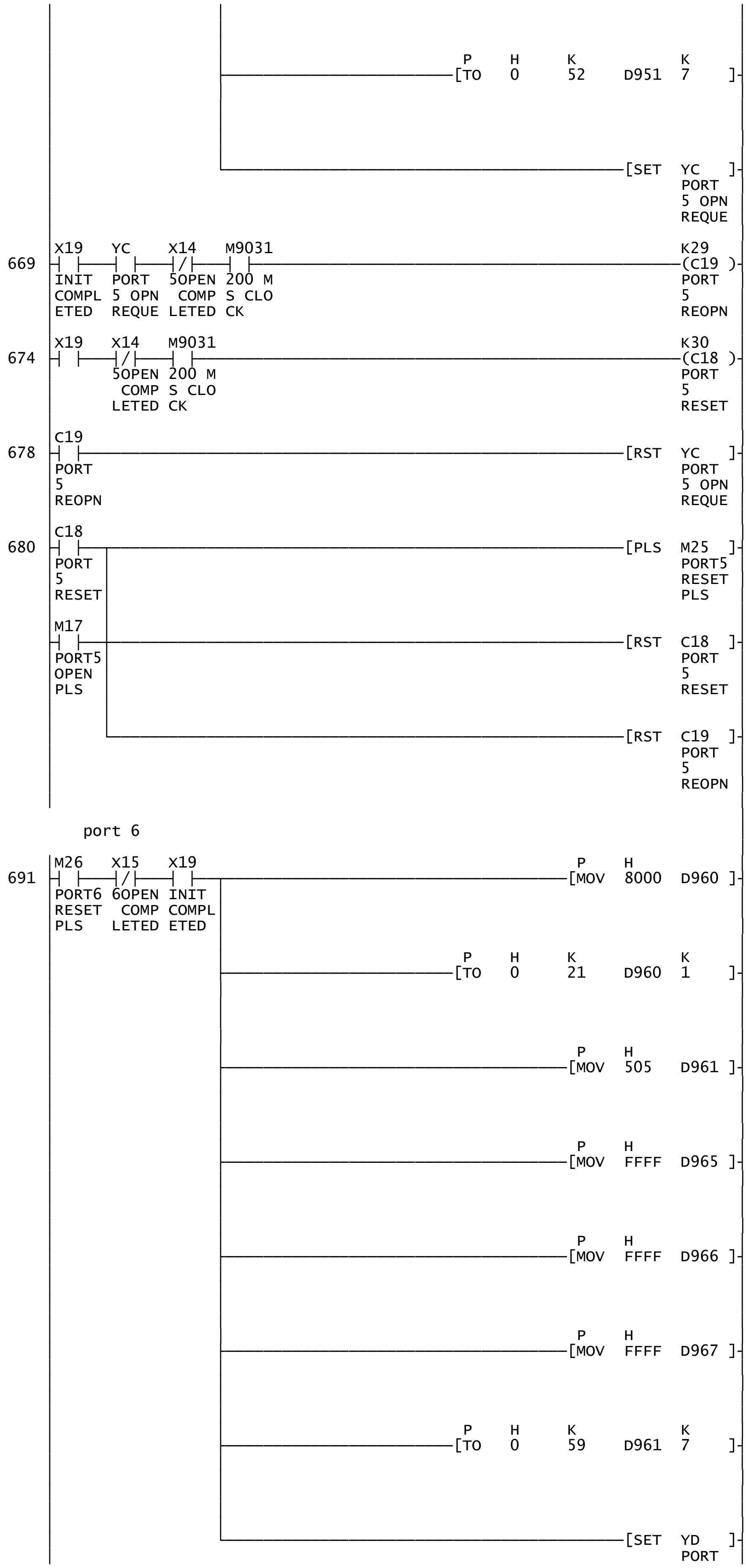
```
             ERROR  ERROR                                        4 OPN
             CODE                                                REQUE
      M16
     -| |-
      PORT4
      OPEN
      PLS

           K          X18
 276  -[<>  0    D995 ]-| |-------------------------------[RST   YC    ]-
                 PORT5 OPEN                                      PORT
                 ERROR ERROR                                     5 OPN
                 CODE                                            REQUE
      M17
     -| |-
      PORT5
      OPEN
      PLS

           K          X18
 284  -[<>  0    D996 ]-| |-------------------------------[RST   YD    ]-
                 PORT6 OPEN                                      PORT
                 ERROR ERROR                                     6 OPN
                 CODE                                            REQUE
      M18
     -| |-
      PORT6
      OPEN
      PLS

           K          X18
 292  -[<>  0    D997 ]-| |-------------------------------[RST   YE    ]-
                 PORT7 OPEN                                      PORT
                 ERROR ERROR                                     7 OPN
                 CODE                                            REQUE
      M19
     -| |-
      PORT7
      OPEN
      PLS

           K          X18
 300  -[<>  0    D998 ]-| |-------------------------------[RST   YF    ]-
                 PORT8 OPEN                                      PORT
                 ERROR ERROR                                     8 OPN
                 CODE                                            REQUE
      M20
     -| |-
      PORT8
      OPEN
      PLS

      X19     X10     X18
 308  -| |-----| |-----|/|--------------------------------[PLS   M11   ]-
      INIT    1OPEN   OPEN                                       Y17
      COMPL    COMP   ERROR                                      LED
      ETED    LETED
              X11
             -| |-
              2OPEN
               COMP
              LETED
              X12
             -| |-
              3OPEN
               COMP
              LETED
              X13
             -| |-
              4OPEN
               COMP
              LETED
              X14
             -| |-
              5OPEN
               COMP
              LETED
              X15
             -| |-
              6OPEN
               COMP
              LETED
              X16
             -| |-
              7OPEN
               COMP
              LETED
```


X17
8OPEN COMP LETED
322
M11
Y17 LED
[SET Y17]
ERROR LED
port 1
324
M21 X10 X19
PORT1 RESET PLS
1OPEN COMP LETED
INIT COMPL ETED
P [MOV H 8000 D910]
OWN_S OCKET
P [TO H 0 K 16 D910 K 1]
OWN_S OCKET
P [MOV H 500 D911]
FORE IGN I P-ADL
P [MOV H FFFF D915]
P [MOV H FFFF D916]
P [MOV H FFFF D917]
P [TO H 0 K 24 D911 K 7]
FORE IGN I P-ADL
[SET Y8]
PORT 1OPEN REQUE
371
X19 Y8 X10 M9031
INIT COMPL ETED
PORT 1OPEN REQUE
1OPEN COMP LETED
200 M S CLO CK
K29 (C11)
PORT 1 REOPE
376
X19 X10 M9031
INIT COMPL ETED
1OPEN COMP LETED
200 M S CLO CK
K30 (C10)
PORT 1 RESET
380
C11
PORT 1 REOPE
[RST Y8]
PORT 1OPEN REQUE
382
C10
PORT 1 RESET
[PLS M21]
PORT1 RESET PLS
M13
PORT1 OPEN PLS
[RST C10]
PORT 1 RESET
[RST C11]
PORT 1 REOPE

```
     port 2

393  M22    X11    X19                                      P     H
     PORT2  2OPEN  INIT                                     [MOV  8000   D920 ]
     RESET   COMP  COMPL
     PLS    LETED  ETED
                                           P     H     K           K
                                           [TO   0     17   D920   1      ]

                                                            P     H
                                                            [MOV  501    D921 ]
                                                                         1
                                                                         ERROR
                                                                         CODE
                                                            P     H
                                                            [MOV  FFFF   D925 ]
                                                                         5
                                                                         ERROR
                                                                         CODE
                                                            P     H
                                                            [MOV  FFFF   D926 ]
                                                                         6
                                                                         ERROR
                                                                         CODE
                                                            P     H
                                                            [MOV  FFFF   D927 ]
                                                                         7
                                                                         ERROR
                                                                         CODE
                                           P     H     K           K
                                           [TO   0     31   D921   7      ]
                                                            1
                                                            ERROR
                                                            CODE

                                                                  [SET   Y9   ]
                                                                         PORT
                                                                         2 OPN
                                                                         REQUE
440  X19    Y9     X11    M9031                                          K29
     INIT   PORT   2OPEN  200 M                                          (C13 )
     COMPL  2 OPN   COMP  S CLO                                          PORT
     ETED   REQUE  LETED  CK                                             2
                                                                         REOPN
445  X19    X11    M9031                                                 K30
     INIT   2OPEN  200 M                                                 (C12 )
     COMPL   COMP  S CLO                                                 PORT
     ETED   LETED  CK                                                    2
                                                                         RESET
449  C13
     PORT                                                         [RST   Y9   ]
     2                                                                   PORT
     REOPN                                                               2 OPN
                                                                         REQUE
451  C12
     PORT                                                         [PLS   M22  ]
     2                                                                   PORT2
     RESET                                                               RESET
                                                                         PLS
     M14
     PORT2                                                        [RST   C12  ]
     OPEN                                                                PORT
     PLS                                                                 2
                                                                         RESET

                                                                  [RST   C13  ]
                                                                         PORT
                                                                         2
                                                                         REOPN

     port 3

484  M23    X12    X19                                      P     H
     PORT3  3OPEN  INIT                                     [MOV  8000   D930 ]
     RESET   COMP  COMPL
     PLS    LETED  ETED
                                           P     H     K           K
                                           [TO   0     18   D930   1      ]
```

```
                                                             P     H
                                                            [MOV   502    D931 ]

                                                             P     H
                                                            [MOV   FFFF   D935 ]

                                                             P     H
                                                            [MOV   FFFF   D936 ]

                                                             P     H
                                                            [MOV   FFFF   D937 ]

                                               P     H      K          K
                                              [TO    0      38   D931  7      ]

                                                                  [SET  YA    ]
                                                                        PORT
                                                                        3 OPN
                                                                        REQUE
      X19   YA    X12   M9031                                           K29
531   ┤ ├───┤ ├───┤/├───┤ ├─────────────────────────────────────────────(C15 )
      INIT  PORT  3OPEN 200 M                                           PORT
      COMPL 3 OPN  COMP S CLO                                           3
      ETED  REQUE LETED CK                                              REOPN
      X19   X12   M9031                                                 K30
536   ┤ ├───┤/├───┤ ├───────────────────────────────────────────────────(C14 )
      INIT  3OPEN 200 M                                                 PORT
      COMPL  COMP S CLO                                                 3
      ETED  LETED CK                                                    RESET
      C15
540   ┤ ├─────────────────────────────────────────────────────────[RST  YA    ]
      PORT                                                              PORT
      3                                                                 3 OPN
      REOPN                                                             REQUE
      C14
542   ┤ ├─────────────────────────────────────────────────────────[PLS  M23   ]
      PORT                                                              PORT3
      3                                                                 RESET
      RESET                                                             PLS
      M15
      ┤ ├─────────────────────────────────────────────────────────[RST  C14   ]
      PORT3                                                             PORT
      OPEN                                                              3
      PLS                                                               RESET

                                                                  [RST  C15   ]
                                                                        PORT
                                                                        3
                                                                        REOPN

         port 4
      M24   X13   X19                                        P     H
553   ┤ ├───┤/├───┤ ├──────────────────────────────────────[MOV   8000   D940 ]
      PORT4 4OPEN INIT
      RESET  COMP COMPL
      PLS   LETED ETED
                                               P     H      K          K
                                              [TO    0      19   D940  1      ]

                                                             P     H
                                                            [MOV   503    D941 ]

                                                             P     H
                                                            [MOV   FFFF   D945 ]
```

```
                                                   P     H
     |                                          -[MOV   FFFF  D946 ]-

                                                   P     H
     |                                          -[MOV   FFFF  D947 ]-

                                          P     H     K           K
     |                                 -[TO    0     45    D941  7     ]-

     |                                               -[SET  YB    ]-
                                                            PORT
                                                            4 OPN
                                                            REQUE

      X19   YB    X13   M9031                               K29
600  -| |---| |---|/|---| |---------------------------------(C17 )-
      INIT  PORT  4OPEN 200 M                               PORT
      COMPL 4 OPN  COMP S CLO                               4
      ETED  REQUE LETED CK                                  REOPN

      X19   X13   M9031                                     K30
605  -| |---|/|---| |---------------------------------------(C16 )-
      INIT  4OPEN 200 M                                     PORT
      COMPL  COMP S CLO                                     4
      ETED  LETED CK                                        RESET

      C17
609  -| |-------------------------------------------[RST   YB    ]-
      PORT                                                  PORT
      4                                                     4 OPN
      REOPN                                                 REQUE

      C16
611  -| |-+-----------------------------------------[PLS   M24   ]-
      PORT |                                                PORT4
      4    |                                                RESET
      RESET|                                                PLS
      M16  |
     -| |--+----------------------------------------[RST   C16   ]-
      PORT4|                                                PORT
      OPEN |                                                4
      PLS  |                                                RESET
           |
           +----------------------------------------[RST   C17   ]-
                                                            PORT
                                                            4
                                                            REOPN

        port 5

      M25   X14   X19                              P     H
622  -| |---|/|---| |-+----------------------------[MOV   8000  D950 ]-
      PORT5 5OPEN INIT |
      RESET  COMP COMPL|
      PLS   LETED ETED |
                       |                  P     H     K           K
                       +----------------[TO    0     20    D950  1     ]-
                       |
                       |                           P     H
                       +-------------------------[MOV   504   D951 ]-
                       |
                       |                           P     H
                       +-------------------------[MOV   FFFF  D955 ]-
                       |
                       |                           P     H
                       +-------------------------[MOV   FFFF  D956 ]-
                       |
                       |                           P     H
                       +-------------------------[MOV   FFFF  D957 ]-
```


P H K K
[TO 0 52 D951 7]
[SET YC]
PORT 5 OPN REQUE
669
X19 YC X14 M9031
INIT COMPL ETED
PORT 5 OPN REQUE
5OPEN COMP LETED
200 M S CLO CK
K29
(C19)
PORT 5 REOPN
674
X19 X14 M9031
5OPEN COMP LETED
200 M S CLO CK
K30
(C18)
PORT 5 RESET
678
C19
PORT 5 REOPN
[RST YC]
PORT 5 OPN REQUE
680
C18
PORT 5 RESET
[PLS M25]
PORT5 RESET PLS
M17
PORT5 OPEN PLS
[RST C18]
PORT 5 RESET
[RST C19]
PORT 5 REOPN
port 6
691
M26 X15 X19
PORT6 RESET PLS
6OPEN COMP LETED
INIT COMPL ETED
P H
[MOV 8000 D960]
P H K K
[TO 0 21 D960 1]
P H
[MOV 505 D961]
P H
[MOV FFFF D965]
P H
[MOV FFFF D966]
P H
[MOV FFFF D967]
P H K K
[TO 0 59 D961 7]
[SET YD]
PORT

```
                                                                        6 OPN
                                                                        REQUE
     X19   YD     X15    M9031                                          K29
738 ─┤ ├───┤ ├────┤/├────┤ ├────────────────────────────────────────────(C21 )─
     INIT  PORT   6OPEN  200 M                                          PORT
     COMPL 6 OPN   COMP  S CLO                                          6
     ETED  REQUE  LETED  CK                                             REOPE

     X19   X15    M9031                                                 K30
743 ─┤ ├───┤/├────┤ ├───────────────────────────────────────────────────(C20 )─
     INIT  6OPEN  200 M                                                 PORT
     COMPL  COMP  S CLO                                                 6
     ETED  LETED  CK                                                    RESET

     C21
747 ─┤ ├──────────────────────────────────────────────────────────[RST  YD    ]─
     PORT                                                               PORT
     6                                                                  6 OPN
     REOPE                                                              REQUE

     C20
749 ─┤ ├──┬───────────────────────────────────────────────────────[PLS  M26   ]─
     PORT │                                                             PORT6
     6    │                                                             RESET
     RESET│                                                             PLS
          │
     M18  │
    ─┤ ├──┼───────────────────────────────────────────────────────[RST  C20   ]─
     PORT6│                                                             PORT
     OPEN │                                                             6
     PLS  │                                                             RESET
          │
          └───────────────────────────────────────────────────────[RST  C21   ]─
                                                                        PORT
                                                                        6
                                                                        REOPE
       port 7

     M27   X16    X19                                              P    H
760 ─┤ ├───┤/├────┤ ├──┬─────────────────────────────────────────[MOV  8000  D970 ]─
     PORT7 7OPEN  INIT │
     RESET  COMP  COMPL│
     PLS   LETED  ETED │
                       │                       P    H     K           K
                       ├───────────────────────[TO  0     22    D970  1     ]─
                       │
                       │                                           P    H
                       ├─────────────────────────────────────────[MOV  506   D971 ]─
                       │
                       │                                           P    H
                       ├─────────────────────────────────────────[MOV  FFFF  D975 ]─
                       │
                       │                                           P    H
                       ├─────────────────────────────────────────[MOV  FFFF  D976 ]─
                       │
                       │                                           P    H
                       ├─────────────────────────────────────────[MOV  FFFF  D977 ]─
                       │
                       │                       P    H     K           K
                       ├───────────────────────[TO  0     66    D971  7     ]─
                       │
                       └──────────────────────────────────────────[SET  YE    ]─
                                                                        PORT
                                                                        7 OPN
                                                                        REQUE
     X19   YE     X16    M9031                                          K29
807 ─┤ ├───┤ ├────┤/├────┤ ├────────────────────────────────────────────(C23 )─
     INIT  PORT   7OPEN  200 M                                          PORT
     COMPL 7 OPN   COMP  S CLO                                          7
     ETED  REQUE  LETED  CK                                             REOPN

     X19   X16    M9031                                                 K30
812 ─┤ ├───┤/├────┤ ├───────────────────────────────────────────────────(C22 )─
     INIT  7OPEN  200 M                                                 PORT
     COMPL  COMP  S CLO                                                 7
```

```
     |ETED  LETED CK                                              RESET  |
     |                                                                   |
     |C23                                                                |
816  |-| |--------------------------------------------------[RST  YE    ]-
     |PORT                                                        PORT   |
     |7                                                           7 OPN  |
     |REOPN                                                       REQUE  |
     |                                                                   |
     |C22                                                                |
818  |-| |-+------------------------------------------------[PLS  M27   ]-
     |PORT  |                                                     PORT7  |
     |7     |                                                     RESET  |
     |RESET |                                                     PLS    |
     |      |                                                            |
     |M19   |                                                            |
     |-| |--+------------------------------------------------[RST  C22   ]-
     |PORT7 |                                                     PORT   |
     |OPEN  |                                                     7      |
     |PLS   |                                                     RESET  |
     |      |                                                            |
     |      +------------------------------------------------[RST  C23   ]-
     |                                                            PORT   |
     |                                                            7      |
     |                                                            REOPN  |

        port 8

     |M28   X17   X19                               P     H              |
829  |-| |--|/|--| |--+-----------------------------[MOV  8000   D980  ]-
     |PORT8 8OPEN INIT |                                                 |
     |RESET  COMP COMPL|                                                 |
     |PLS   LETED ETED |                                                 |
     |                 |                                                 |
     |                 |          P     H     K            K             |
     |                 +----------[TO   0     23    D980   1     ]-
     |                 |                                                 |
     |                 |                            P     H              |
     |                 +----------------------------[MOV  507    D981  ]-
     |                 |                                                 |
     |                 |                            P     H              |
     |                 +----------------------------[MOV  FFFF   D985  ]-
     |                 |                                                 |
     |                 |                            P     H              |
     |                 +----------------------------[MOV  FFFF   D986  ]-
     |                 |                                                 |
     |                 |                            P     H              |
     |                 +----------------------------[MOV  FFFF   D987  ]-
     |                 |                                                 |
     |                 |          P     H     K            K             |
     |                 +----------[TO   0     73    D981   7     ]-
     |                 |                                                 |
     |                 +---------------------------------[SET  YF    ]-
     |                                                            PORT   |
     |                                                            8 OPN  |
     |                                                            REQUE  |
     |X19   YF    X17   M9031                                 K29      |
876  |-| |--| |--|/|--| |-------------------------------------(C25  )-
     |INIT  PORT  8OPEN 200 M                                 PORT     |
     |COMPL 8 OPN  COMP S CLO                                 8        |
     |ETED  REQUE LETED CK                                    REOPN    |
     |X19   X17   M9031                                       K30      |
881  |-| |--|/|--| |-------------------------------------------(C24  )-
     |INIT  8OPEN 200 M                                       PORT     |
     |COMPL  COMP S CLO                                       8        |
     |ETED  LETED CK                                          RESET    |
     |C25                                                              |
885  |-| |--------------------------------------------------[RST  YF    ]-
     |PORT                                                        PORT   |
     |8                                                           8 OPN  |
     |REOPN                                                       REQUE  |
     |C24                                                                |
887  |-| |-+------------------------------------------------[PLS  M28   ]-
     |PORT  |                                                     PORT8  |
     |8     |                                                     RESET  |
```

```
    |RESET|                                                          PLS   |
    |     |                                                                |
    |M20  |                                                                |
    |-| |-+------------------------------------------------[RST    C24   ]-|
    |PORT8|                                                        PORT    |
    |OPEN |                                                        8       |
    |PLS  |                                                        RESET   |
    |     |                                                                |
    |     +------------------------------------------------[RST    C25   ]-|
    |                                                              PORT    |
    |                                                              8       |
    |                                                              REOPN   |
       port 1 hb
    |T1     M1                                                     K1200   |
898 |-|/|----|/|---------------------------------------------------(T1   )-|
    |PORT1 PORT1                                                   PORT1   |
    |OPEN  HEART                                                   OPEN    |
    |RESET BIT                                                     RESET   |
    |M1                                                                    |
901 |-| |--------------------------------------------------[RST    M1    ]-|
    |PORT1                                                         PORT1   |
    |HEART                                                         HEART   |
    |BIT                                                           BIT     |
    |T1                                                                    |
903 |-| |--------------------------------------------------[RST    Y8    ]-|
    |PORT1                                                         P0RT    |
    |OPEN                                                          1OPEN   |
    |RESET                                                         REQUE   |
       port 2 hb
    |T2     M2                                                     K1200   |
905 |-|/|----|/|---------------------------------------------------(T2   )-|
    |PORT2 PORT2                                                   PORT2   |
    |OPEN  HEART                                                   OPEN    |
    |RESET BIT                                                     RESET   |
    |M2                                                                    |
908 |-| |--------------------------------------------------[RST    M2    ]-|
    |PORT2                                                         PORT2   |
    |HEART                                                         HEART   |
    |BIT                                                           BIT     |
    |T2                                                                    |
910 |-| |--------------------------------------------------[RST    Y9    ]-|
    |PORT2                                                         PORT    |
    |OPEN                                                          2 OPN   |
    |RESET                                                         REQUE   |
       port 3 hb
    |T3     M3                                                     K1200   |
912 |-|/|----|/|---------------------------------------------------(T3   )-|
    |PORT3 PORT3                                                   PORT3   |
    |OPEN  HEART                                                   OPEN    |
    |RESET BIT                                                     RESET   |
    |M3                                                                    |
915 |-| |--------------------------------------------------[RST    M3    ]-|
    |PORT3                                                         PORT3   |
    |HEART                                                         HEART   |
    |BIT                                                           BIT     |
    |T3                                                                    |
917 |-| |--------------------------------------------------[RST    YA    ]-|
    |PORT3                                                         PORT    |
    |OPEN                                                          3 OPN   |
    |RESET                                                         REQUE   |
       port 4 hb
    |T4     M4                                                     K1200   |
919 |-|/|----|/|---------------------------------------------------(T4   )-|
    |PORT4 PORT4                                                   PORT4   |
    |OPEN  HEART                                                   OPEN    |
    |RESET BIT                                                     RESET   |
    |M4                                                                    |
922 |-| |--------------------------------------------------[RST    M4    ]-|
    |PORT4                                                         PORT4   |
    |HEART                                                         HEART   |
    |BIT                                                           BIT     |
    |T4                                                                    |
924 |-| |--------------------------------------------------[RST    YB    ]-|
    |PORT4                                                         PORT    |
    |OPEN                                                          4 OPN   |
    |RESET                                                         REQUE   |
       port 5 hb
```

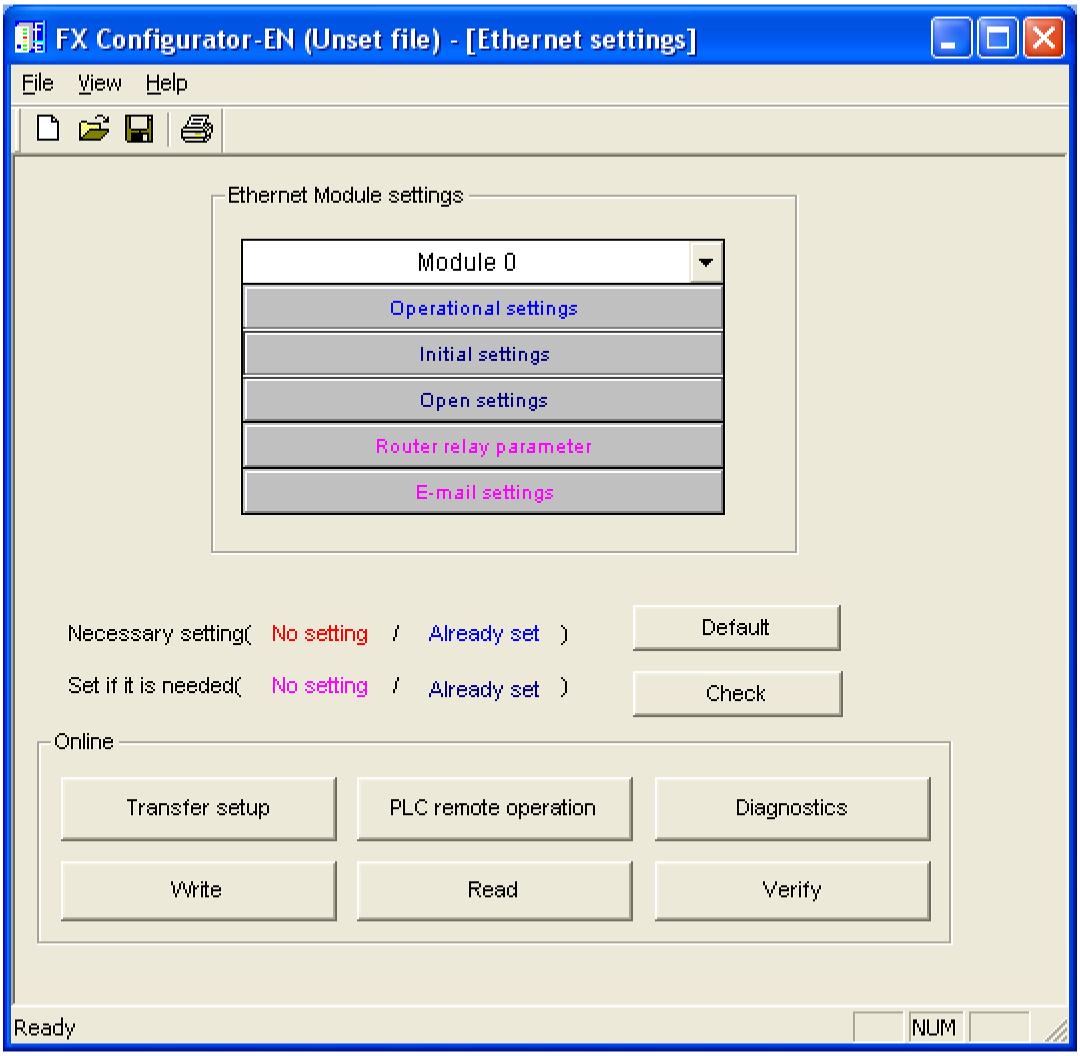
```
    |T5    M5                                                              K1200
926 +|/|---|/|-------------------------------------------------------------(T5   )-|
    |PORT5 PORT5                                                           PORT5   |
    |OPEN  HEART                                                           OPEN    |
    |RESET BIT                                                             RESET   |
    |                                                                              |
    |M5                                                                            |
929 +| |-----------------------------------------------------------[RST    M5    ]-|
    |PORT5                                                                 PORT5   |
    |HEART                                                                 HEART   |
    |BIT                                                                   BIT     |
    |                                                                              |
    |T5                                                                            |
931 +| |-----------------------------------------------------------[RST    YC    ]-|
    |PORT5                                                                 PORT    |
    |OPEN                                                                  5 OPN   |
    |RESET                                                                 REQUE   |
    |                                                                              |

       port 6 hb

    |T6    M6                                                              K1200
933 +|/|---|/|-------------------------------------------------------------(T6   )-|
    |PORT6 PORT6                                                           PORT6   |
    |OPEN  HEART                                                           OPEN    |
    |RESET BIT                                                             RESET   |
    |                                                                              |
    |M6                                                                            |
936 +| |-----------------------------------------------------------[RST    M6    ]-|
    |PORT6                                                                 PORT6   |
    |HEART                                                                 HEART   |
    |BIT                                                                   BIT     |
    |                                                                              |
    |T6                                                                            |
938 +| |-----------------------------------------------------------[RST    YD    ]-|
    |PORT6                                                                 PORT    |
    |OPEN                                                                  6 OPN   |
    |RESET                                                                 REQUE   |
    |                                                                              |

       port 7 hb

    |T7    M7                                                              K1200
940 +|/|---|/|-------------------------------------------------------------(T7   )-|
    |PORT7 PORT7                                                           PORT7   |
    |OPEN  HEART                                                           OPEN    |
    |RESET BIT                                                             RESET   |
    |                                                                              |
    |M7                                                                            |
943 +| |-----------------------------------------------------------[RST    M7    ]-|
    |PORT7                                                                 PORT7   |
    |HEART                                                                 HEART   |
    |BIT                                                                   BIT     |
    |                                                                              |
    |T7                                                                            |
945 +| |-----------------------------------------------------------[RST    YD    ]-|
    |PORT7                                                                 PORT    |
    |OPEN                                                                  6 OPN   |
    |RESET                                                                 REQUE   |
    |                                                                              |

       port 8 hb

    |T8    M8                                                              K1200
947 +|/|---|/|-------------------------------------------------------------(T8   )-|
    |PORT8 PORT8                                                           PORT8   |
    |OPEN  HEART                                                           OPEN    |
    |RESET BIT                                                             RESET   |
    |                                                                              |
    |M8                                                                            |
950 +| |-----------------------------------------------------------[RST    M8    ]-|
    |PORT8                                                                 PORT8   |
    |HEART                                                                 HEART   |
    |BIT                                                                   BIT     |
    |                                                                              |
    |T8                                                                            |
952 +| |-----------------------------------------------------------[RST    YF    ]-|
    |PORT8                                                                 PORT    |
    |OPEN                                                                  8 OPN   |
    |RESET                                                                 REQUE   |
   **************************************************
   *                    COMS END                    *
   **************************************************
```

Appendix B

How to configure the Fx3U Ethernet module for communication to Adroit

Step 1:

When starting up the FX Configurator-EN software you will be represented with the following screen. Under Ethernet Module Settings there is a drop down box to select which module your Ethernet card is coneected as. “Module 0” selection means it is the first card after the CPU.

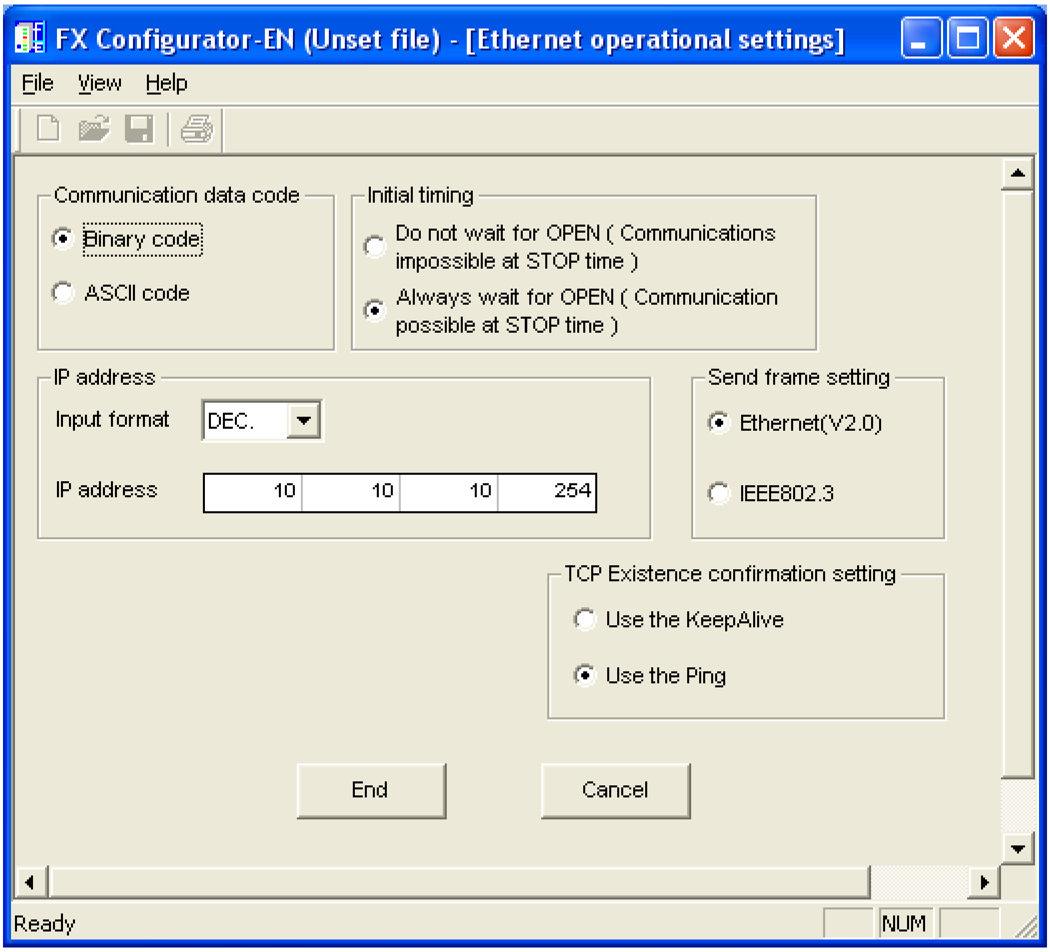


Step 2:

Click on the “Operational Settings” button to get the next screen.

In the operational settings screen, the settings should look exactly as below, except for your IP address that might differ. This will be your IP address you want to configure the Ethernet card for.

Once completed press “END” to take you back to the main screen.

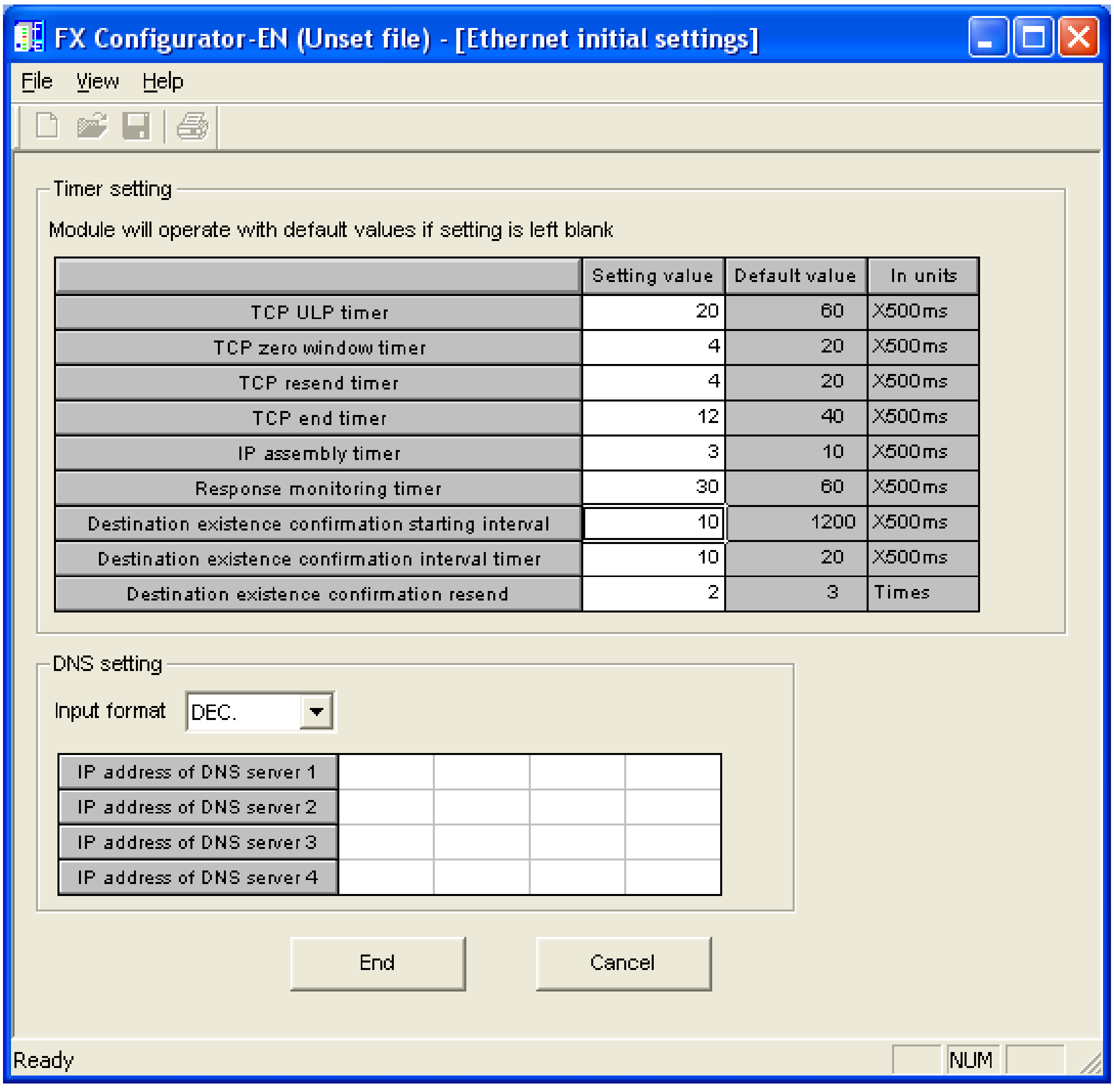


Step 3:

Click on the "Initial Settings" button to get to the next screen.

In the initial settings screen, the settings should look exactly as below.

Once completed press "END" to take you back to the main screen.

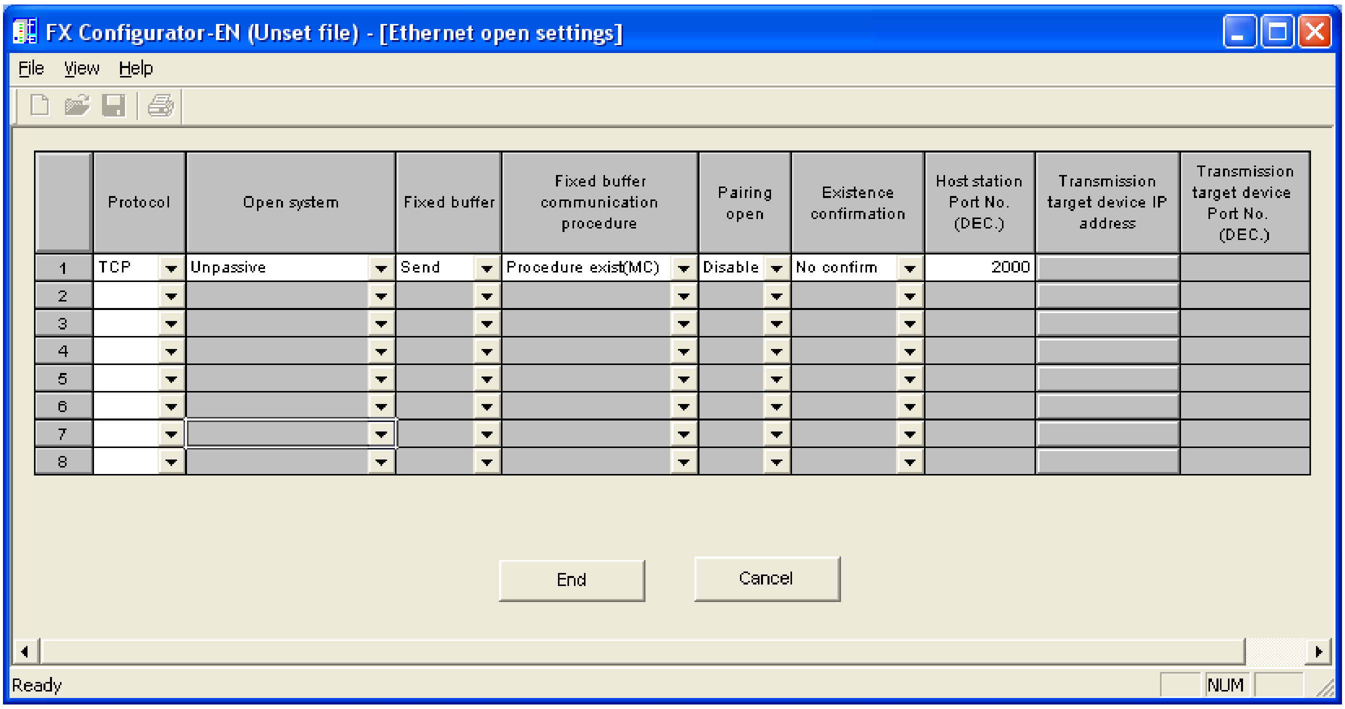


Step 4:

Click on the “Open Settings” button to get to the next screen.

In the open settings screen, the settings should look exactly as below, except for your Port which you can configure to be what you require the card port to be.

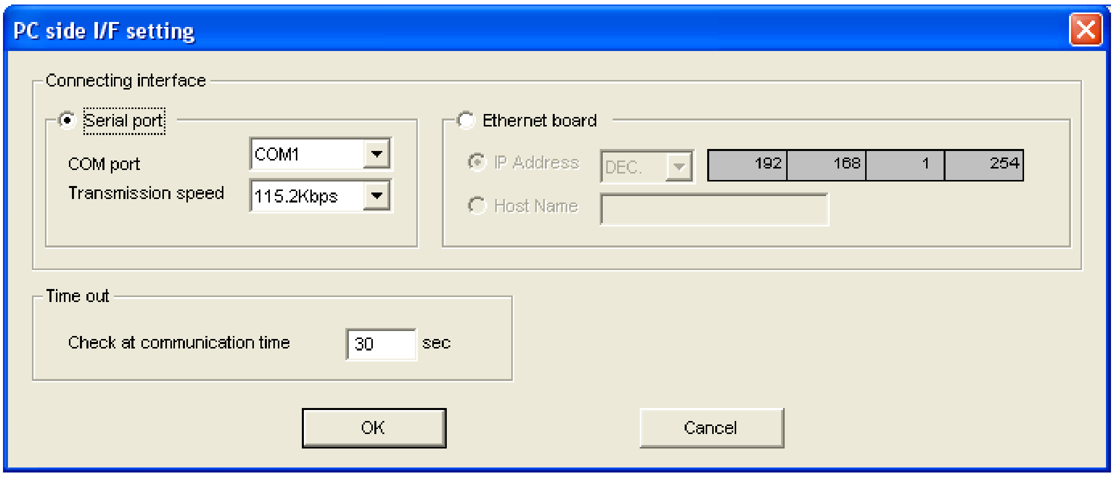
Once completed press “END” to take you back to the main screen.



Step 5:

Click on the “Transfer Setup” button to get to the next screen.

In this screen, you select which method to use download the setup to the PLC. As we only configured the Ethernet module now, we will select “Serial Port” and press ok to take you back to the main screen. You will require the “RED” mitshubishi beijer SC-09 serial cable to do the download.

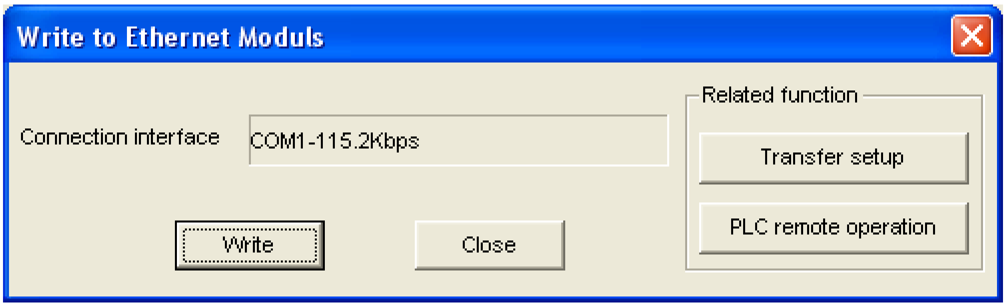


Step 6:

Click on the “Write” button to get to the next screen.

In this screen, you just click “Write” and it will start to download the setup to the PLC.

It should return with a message box that says “completed”. Press OK and close the whole software package. Your Fx3U Ethernet card will no be set.


Write to Ethernet Moduls
Related function
Connection interface
COM1-115.2Kbps
Transfer setup
PLC remote operation
Write
Close

Revision Information

Date	Changes
13/06/02	PLC program listing added
25/11/02	Q-Series setup covered + (heartbeat disable-able)
28/03/03	Annunciators “F” supported
7/11/03	Address notation style table added
24/05/05	Ladder program rung at 674 should reference coil X19, not coil X9. Driver rev 101 drops support for Q, and fixes a bug which prevented proper A-series comms.
25/08/05	Watchdog heat-beat troubleshooting paragraph added
06/10/06	Added Fx3U Ethernet card setup information
14/06/2013	Added Test Read to Dialog